

PUPIL NAME: \_\_\_\_\_ CLASS: \_\_\_\_\_ TEACHER: \_\_\_\_\_

KEY TOPIC 4 • C1900–PRESENT

Medicine in Modern Britain

DNA, High-Tech Diagnosis, The NHS, Penicillin, and The War on Lung Cancer



Primary Archival Source: *The Dawn of Free Healthcare: The 1948 National Health Service Leaflet*      Accession Shelfmark: ARCH-MED-1948

| COURSE SPECIFICATION • KEY ENQUIRY SEQUENCE                                                                                                                                                                                              | LEARNT                   | REVISED                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|
| <b>Key Topic 4.1: Ideas on Causes: Genetics, DNA &amp; The Human Genome Project</b><br><i>How did Rosalind Franklin’s Photograph 51, Watson &amp; Crick, and genomic sequencing unlock hereditary disease?</i>                           | <input type="checkbox"/> | <input type="checkbox"/> |
| <b>Key Topic 4.2: Lifestyle Factors &amp; The Technological Revolution in Diagnosis</b><br><i>How did the shift to chronic lifestyle illnesses, Röntgen’s X-rays, and Hounsfield’s CT scanner eliminate exploratory surgery?</i>         | <input type="checkbox"/> | <input type="checkbox"/> |
| <b>Key Topic 4.3: Magic Bullets, High-Tech Treatments &amp; The Birth of the NHS</b><br><i>How did Salvarsan 606, Prontosil, and Aneurin Bevan’s 1948 NHS create free healthcare for every British citizen?</i>                          | <input type="checkbox"/> | <input type="checkbox"/> |
| <b>Key Topic 4.4: Case Study 1: The Antibiotic Revolution: Fleming, Florey &amp; Chain</b><br><i>How did Fleming’s 1928 discovery, the Oxford team’s purification, and American deep-tank vats deliver 2.3m doses for D-Day?</i>         | <input type="checkbox"/> | <input type="checkbox"/> |
| <b>Key Topic 4.5: Case Study 2: Public Health &amp; The Fight Against Lung Cancer</b><br><i>How did Doll &amp; Hill prove the tobacco link, and how did ad bans, the 2007 public smoking ban, and plain packaging end laissez-faire?</i> | <input type="checkbox"/> | <input type="checkbox"/> |

## Living Unit Timeline: Modern Medicine (Part I: 1900–1940)

## CHRONOLOGICAL FRAMEWORK • MAGIC BULLETS & EARLY ANTIBIOTICS

1901

## LANDSTEINER DISCOVERS BLOOD GROUPS

Karl Landsteiner identifies A, B, and O blood groups, transforming blood transfusions from fatal gambles into predictable surgical lifelines.

1909

**EHRlich & HATA DISCOVER SALVARISAN 606**

Paul Ehrlich and Sahachiro Hata discover arsphenamine (compound 606), humanity's first synthetic chemical 'magic bullet' curing syphilis.

# 1928

## ALEXANDER FLEMING DISCOVERS PENICILLIN

Fleming observes a clear zone of destroyed staphylococci around *Penicillium notatum* mold at St Mary's Hospital, Paddington, London.

1932

## DOMAGK DISCOVERS PRONTOSIL

Gerhard Domagk proves the red dye Prontosil cures streptococcal blood poisoning, establishing the sulfonamide class of antibacterial drugs.

## • LIVING TIMELINE SKETCHPAD & CAUSAL SYNTHESIS (1900–1940)

Sketch Ehrlich's Salvarsan vial, Domagk's Prontosil, or Fleming's contaminated staphylococcus petri dish.

## Living Unit Timeline: Modern Medicine (Part II: 1940–present)

## MASS PRODUCTION OF PENICILLIN FOR D-DAY

Following Florey and Heatley's US mission to Peoria, deep-tank fermentation yields 2.3 million doses of penicillin for the Normandy landings.

## ANEURIN BEVAN LAUNCHES THE NHS

On 5 July 1948, the National Health Service opens at Park Hospital, Manchester, providing comprehensive healthcare free at the point of delivery.

## WATSON & CRICK DISCOVER DNA DOUBLE HELIX

Using Rosalind Franklin's Photograph 51, James Watson and Francis Crick deduce complementary base pairing, unlocking the molecular secret of life.

### CT SCANNER (1971) & HUMAN GENOME (2003)

Hounsfield's EMI CT scanner eliminates exploratory surgery; the Human Genome Project maps all 3 billion chemical base pairs in human DNA.

• **LIVING TIMELINE SKETCHPAD & CAUSAL SYNTHESIS (1940–PRESENT)** *Sketch the 1948 NHS leaflet, Franklin's Photograph 51, Watson & Crick's double helix, or Hounsfield's CT scanner.*

## • 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [   / 10 ]

1. Which ancient Greek physician created the Theory of the Four Humours?

3. In what year did Andreas Vesalius publish *De Humani Corporis Fabrica*?

5. Which English doctor was nicknamed the 'English Hippocrates' for bedside observation?

7. Which French chemist disproved spontaneous generation and published Germ Theory in 1861?

9. What anaesthetic did James Simpson discover in Edinburgh in 1847?

2. Which Roman doctor introduced the Theory of Opposites?

4. What did William Harvey prove about blood flow in 1628?

6. What vaccine did Edward Jenner develop in 1796 using cowpox matter?

8. Which German bacteriologist isolated the bacteria causing anthrax (1876) and tuberculosis (1882)?

10. What 1875 Act compelled local councils in Britain to provide clean water and sewers?

## • KEY VOCABULARY TASK

TERMINOLOGY

Define **DNA (Deoxyribonucleic Acid)** and explain how the **Human Genome Project** transformed disease prediction:

## • QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which ideas about the cause of disease in 1900 were similar to ideas in 1880 in Britain. [4 marks]

*Hint: Focus on the universal acceptance of Germ Theory and the belief that all human illnesses were caused exclusively by external microscopic bacteria.*

**Stems:** One way ideas about causes were similar was the reliance on Germ Theory... • In 1880, Koch had proven that... • Similarly, in 1900, doctors still assumed that all diseases...

## • QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which scientific methods for investigating disease causes in 1990 were different from methods in 1880. [4 marks]

*Hint: Contrast the Human Genome Project's computerized biochemical DNA sequencing with Robert Koch's manual microscopic staining with aniline dyes.*

**Stems:** One way methods differed was the use of molecular genetics... • In 1880, Koch relied on... • In contrast, by 1990, geneticists used automated computers to...

## EXTENDED WRITING ASSESSMENT

**Stimulus:**

## The Human Genome Project (1990–2003)

(You must also use information of your own)

## 1. FRANKLIN & X-RAY CRYSTALLOGRAPHY

## 2. WATSON & CRICK'S DOUBLE HELIX

### 3. THE HUMAN GENOME PROJECT (1990–2003)

### ANALYTICAL CONNECTIVES:

**DOMAIN VOCABULARY BANK:**

DNA • Rosalind Franklin • Maurice Wilkins • Photograph 51 (1952) • X-ray crystallography • James Watson • Francis Crick • double helix (1953) • base pairing (A-T, C-G) • Human Genome Project (1990–2003) • 3 billion base pairs • BRCA1/2 • gene therapy

## SYNTHESIS

*Mapping all 3 billion human DNA base pairs took 13 years; now geneticists can read our biological instruction manual.*

### • 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [   / 10 ]

1. In what year did Wilhelm Röntgen discover X-rays?

2. Who operated 20 mobile X-ray vehicles ("petites Curies") during WWI?

3. What Scottish surgeon used carbolic acid spray in 1865 to prevent wound sepsis?

4. What fatal epidemic struck London in 1854, centered on Broad Street?

5. What doctor used a spot map to identify the contaminated Broad Street pump in 1854?

6. What chemical base does Adenine pair with in the DNA double helix?

7. What chemical base does Cytosine pair with in the DNA double helix?

8. Who captured Photograph 51 in May 1952 using X-ray crystallography?

9. What major project mapped all 3 billion base pairs in human DNA (1990–2003)?

10. What term describes the shift from infectious diseases to chronic lifestyle conditions?

### • KEY VOCABULARY TASK

TERMINOLOGY

Explain the fundamental diagnostic difference between a **Standard 2D X-ray** (1895) and a **Computed Tomography (CT) Scan** (1971):

### • QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which diagnosis in 1900 was similar to diagnosis in 1700 in Britain. [4 marks]

*Hint: Focus on doctors relying on external bedside observations (taking pulse, inspecting tongue, smelling urine) and blind exploratory surgery.*

**Stems:** One way diagnosis was similar was the reliance on external bedside guesswork... • In 1700, Sydenham taught doctors to... • Similarly, in 1900, before modern scanning, physicians still had to...

### • QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which diagnosis in 2000 was different from diagnosis in 1900 in Britain. [4 marks]

*Hint: Contrast high-tech non-invasive imaging (CT, MRI, ultrasound) and automated blood labs with external symptom observation and exploratory surgery.*

**Stems:** One way diagnosis differed was the use of non-invasive 3D internal imaging... • In 1900, doctors had to... • In contrast, by 2000, CT and MRI scanners allowed clinicians to...

**Stimulus:** Wilhelm Röntgen's X-rays (1895)      The 2007 smoking ban in public places      (You must also use information of your own)

Wilhelm Röntgen (1895) • X-rays • Marie Curie • mobile radiology • Godfrey Hounsfield • CT scanner (1971) • Atkinson Morley Hospital • MRI • ultrasound • endoscope • lifestyle factors • cardiovascular disease • type 2 diabetes • epidemiological transition

## 7/14

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [   / 10 ]

1. What term did Paul Ehrlich use for chemical compounds that destroy bacteria without harming patient cells?

2. What disease was cured by Salvarsan 606 in 1909?

3. Which Japanese bacteriologist assisted Paul Ehrlich in discovering Salvarsan 606?

4. Who discovered that the red dye Prontosil cured streptococcal blood poisoning in 1932?

5. What active antibacterial chemical group was found inside Prontosil?

6. Who authored the 1942 social report identifying the "Five Giant Evils" in Britain?

7. Who was appointed Minister of Health in 1945 to create the National Health Service?

8. On what exact date was the NHS officially launched in Britain?

9. Which medical organization fiercely resisted state-employed doctors under the NHS?

10. Which British engineer invented the CT scanner at EMI laboratories in 1971?

• KEY VOCABULARY TASK

TERMINOLOGY

Define a **Magic Bullet** and explain how **Salvarsan 606** fundamentally differed from older chemical antiseptics:

• QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which healthcare access for working-class families before 1948 was similar to the nineteenth century. [4 marks]

*Hint: Focus on exclusion from state healthcare, reliance on private doctor fees, charity hospitals, and the dread of medical debt.*

**Stems:** One way access was similar was the financial barrier to medical care... • In the nineteenth century, poor families... • Similarly, before 1948, women and children excluded from National Insurance had to...

• QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which hospital care after 1948 was different from hospital care in the 1930s in Britain. [4 marks]

*Hint: Contrast complete universal care free at the point of delivery funded by taxation with bankrupt voluntary hospitals and means-tested fees.*

**Stems:** One way hospital care differed was the abolition of fees at the point of treatment... • In the 1930s, voluntary hospitals... • In contrast, by 1948, the NHS provided free treatment funded by...



**Stimulus:** The National Health Service (1948)      The discovery of Salvarsan 606 (1909)      *(You must also use information of your own)*

|                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                       |                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>1. THE NHS REVOLUTION (1948)</b><br/>         Explain the 1942 Beveridge Report, Bevan defeating BMA opposition, healthcare free at delivery funded by central taxation, and abolishing medical poverty.</p>                                                                                                                                  | <p><b>2. PHARMACOLOGICAL MAGIC BULLETS</b><br/>         Explain Ehrlich &amp; Hata's Salvarsan 606 (1909 for syphilis), Domagk's Prontosil (1932 for blood poisoning), providing the cures that hospitals needed.</p> | <p><b>3. CRITERIA &amp; FINANCIAL STRAIN</b><br/>         Evaluate economic strain (1951 prescription charges), Bevan's compromise with consultants, and modern high-tech therapy costs (dialysis, transplants).</p> |
| <p><b>ANALYTICAL CONNECTIVES:</b><br/> <i>On the one hand, the NHS was a unique institutional turning point because...</i><br/> <i>• In particular, it guaranteed... • On the other hand, healthcare access relied on pharmacological breakthroughs like... • Furthermore, economic limits forced... • Overall, I judge that...</i></p>             |                                                                                                                                                                                                                       |                                                                                                                                                                                                                      |
| <p><b>DOMAIN VOCABULARY BANK:</b><br/>         Zauberkegel (magic bullet) • Paul Ehrlich • Sahachiro Hata • Salvarsan 606 (1909) • Treponema pallidum • Gerhard Domagk • Prontosil (1932) • sulphonamides • William Beveridge (1942) • Five Giants • Aneurin Bevan • BMA opposition • Park Hospital Manchester (5 July 1948) • central taxation</p> |                                                                                                                                                                                                                       |                                                                                                                                                                                                                      |

## SYNTHESIS

*Bevan overcame doctor opposition by "stuffing their mouths with gold," launching the NHS on 5 July 1948.*

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [     / 10 ]

1. In what year did Alexander Fleming discover penicillin at St Mary's Hospital?

2. What type of bacteria was being cultured when penicillin was discovered?

3. What was the scientific name of the mold that contaminated Fleming's dish?

4. Which two Oxford scientists reopened penicillin research in 1938?

5. What technician built improvised laboratory apparatus using bedpans at Oxford?

6. What was the name of the Oxford policeman who was the first human patient treated with penicillin in 1941?

7. Why did Albert Alexander die despite penicillin clearing his infection?

8. In what US city did scientists discover that corn steep liquor boosted penicillin yields?

9. How many doses of penicillin were manufactured in time for D-Day in June 1944?

10. What dangerous consequence of overusing antibiotics did Fleming warn about in 1945?

• KEY VOCABULARY TASK

TERMINOLOGY

Explain the fundamental chemical difference between an **Antiseptic** (Lister) and an **Antibiotic** (Fleming, Florey & Chain):

• QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which Alexander Fleming's discovery of penicillin in 1928 was similar to Edward Jenner's discovery of the smallpox vaccine in 1796. [4 marks]

*Hint: Focus on both discoveries relying on serendipitous/accidental observation of natural living organisms protecting against disease.*

**Stems:** One way both discoveries were similar was the role of serendipitous observation... • In 1796, Jenner observed that... • Similarly, in 1928, Fleming noticed that a stray mold...

• QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which the mass production of penicillin in 1944 was different from early production in 1940. [4 marks]

*Hint: Contrast 10,000-gallon American deep-tank fermentation vats producing millions of doses with the Oxford team culturing mold in ceramic bedpans.*

**Stems:** One way production differed was the industrial scale and technology... • In 1940, the Oxford team had to... • In contrast, by 1944, US chemical plants utilized...

## Question 4: Explain Why [12 marks • 15 mins]

## EXTENDED WRITING ASSESSMENT

**Explain why penicillin was successfully developed and mass-produced in the period 1928–1945. [12 marks]**

**Stimulus:** Alexander Fleming's discovery (1928) The impact of the Second World War (You must also use information of your own)

### STRUCTURE STRIP • 3-PARAGRAPH ANALYTICAL PLAN

#### 1. FLEMING'S 1928 OBSERVATION

Explain St Mary's Hospital, the contaminated staphylococcus plate, the clear halo of destroyed bacteria, and Fleming publishing in 1929.

#### 2. THE OXFORD TEAM (1938–41)

Explain Florey, Chain & Heatley, freeze-drying extraction, the 1940 mice trial, and the tragic 1941 Albert Alexander clinical proof.

#### 3. US WARTIME MASS PRODUCTION

Explain the 1941 Peoria mission, corn steep liquor, cantaloupe strain, deep-tank vats, and 2.3 million doses available for D-Day (1944).

#### ANALYTICAL CONNECTIVES:

*A primary catalyst was Fleming's acute observation in 1928... • In particular, he noticed... • Building directly on this, Florey and Chain solved the purification crisis by... • Most decisively, the Second World War prompted US industrialization because... • Therefore...*

#### DOMAIN VOCABULARY BANK:

Alexander Fleming • St Mary's Hospital (1928) • Staphylococcus • Penicillium notatum • Howard Florey • Ernst Chain • Norman Heatley • freeze-drying • 1940 mouse trial • Albert Alexander (1941) • Peoria Illinois (1941) • corn steep liquor • Penicillium chrysogenum • deep-tank fermentation • D-Day (1944) • 2.3 million doses

#### • LIVING TIMELINE MISSION

#### SYNTHESIS

Turn to Pages 2–3 (Key Topic 4.4). In the drawing box, sketch Fleming's contaminated petri dish and an American 10,000-gallon deep-tank vat. Annotate how penicillin saved Allied troops on D-Day!

### • 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [   / 10 ]

1. Who were the two British epidemiologists who proved that smoking causes lung cancer in 1950?

2. What was the long-term study of over 40,000 doctors launched in 1951 called?

3. Why is lung cancer particularly difficult to diagnose in its early stages?

4. What type of diagnostic scan uses low-dose radiation to detect small lung tumors early?

5. What surgical procedure involves removing an entire cancerous lobe of a lung?

6. What modern cancer treatment uses drugs to train the body's own immune T-cells to attack tumors?

7. In what year was cigarette advertising completely banned on British television?

8. In what year was smoking banned in all enclosed public places and workplaces in England?

9. What standardized color (Pantone 448 C) was mandated for UK cigarette packaging in 2016?

10. Approximately what percentage of adult British men smoked in 1950 compared to under 13% today?

### • KEY VOCABULARY TASK

TERMINOLOGY

Explain how **Plain Packaging (2016)** and the **Health Act 2007** represent the complete, permanent end of government **Laissez-Faire**:

### • QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which the government reaction to lung cancer after 1965 was similar to the Public Health Act of 1875. [4 marks]

*Hint: Focus on the total abandonment of laissez-faire in favor of compulsory national legislation protecting citizens from environmental health hazards.*

**Stems:** One way government action was similar was the rejection of laissez-faire... • Under the 1875 Act, Parliament compelled councils to... • Similarly, after 1965, the government used compulsory laws to ban advertising and...

### • QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which treatments for lung cancer in the 2000s were different from treatments in 1950 in Britain. [4 marks]

*Hint: Contrast modern targeted immunotherapy, robotic keyhole lobectomy, and precision radiotherapy with rudimentary palliative care and high operative mortality.*

**Stems:** One way treatments differed was the development of high-tech oncology... • In 1950, doctors could only offer... • In contrast, by the 2000s, oncologists used targeted immunotherapy and robotic lobectomies to...

*(You must also use information of your own)*

lung cancer • cigarette smoking • Richard Doll & Austin Bradford Hill (1950) • British Doctors Study (1951) • low-dose helical CT scan • bronchoscopy • lobectomy (VATS) • stereotactic radiotherapy • chemotherapy • immunotherapy • TV advertising ban (1965) • Health Act 2007 (public ban) • plain packaging (2016) • smoke-free generation

## 13/14

| #                                   | Enquiry Lesson Title           | 4-Mark Task               | Extended Writing          | Total Score     |
|-------------------------------------|--------------------------------|---------------------------|---------------------------|-----------------|
|                                     | KT4.1: DNA & Human Genome      | Q3 Diff: [ / 4 ]          | Q4 Why: [ / 12 ]          | [ / 16 ]        |
|                                     | KT4.2: Lifestyle & Diagnosis   | Q3 Sim: [ / 4 ]           | Q5/6 Essay: [ / 20 ]      | [ / 24 ]        |
|                                     | KT4.3: Magic Bullets & The NHS | Q3 Sim: [ / 4 ]           | Q5/6 Essay: [ / 20 ]      | [ / 24 ]        |
|                                     | KT4.4: Penicillin Case Study   | Q3 Diff: [ / 4 ]          | Q4 Why: [ / 12 ]          | [ / 16 ]        |
|                                     | KT4.5: Lung Cancer Case Study  | Q3 Sim: [ / 4 ]           | Q5/6 Essay: [ / 20 ]      | [ / 24 ]        |
| <b>CUMULATIVE ASSESSMENT TOTALS</b> |                                | <b>Q3 Total: [ / 20 ]</b> | <b>Extended: [ / 76 ]</b> | <b>[ / 96 ]</b> |

### TEACHER FORMATIVE ASSESSMENT • FEEDBACK

Effort: [ 1 • 2 • 3 • 4 • 5 ]

#### WHAT WENT WELL (WWW):

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#### EVEN BETTER IF (EBI):

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Teacher Signature: \_\_\_\_\_

Date: \_\_\_\_\_

### INTERACTIVE DIGITAL QUIZZING HUB • SCAN FOR INSTANT RETRIEVAL PRACTICE

ONLINE RECALL

|                                                                                                                                                                               |                                                                                                                                                                               |                                                                                                                                                                                    |                                                                                                                                                                              |                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>KT4.1</b><br/>DNA &amp; Genetics</p>  <p>SCAN TO QUIZ</p> <p>Best Score: [ / 8 ]</p> | <p><b>KT4.2</b><br/>Diagnosis &amp; CT</p>  <p>SCAN TO QUIZ</p> <p>Best Score: [ / 8 ]</p> | <p><b>KT4.3</b><br/>Magic Bullets &amp; NHS</p>  <p>SCAN TO QUIZ</p> <p>Best Score: [ / 8 ]</p> | <p><b>KT4.4</b><br/>Penicillin Study</p>  <p>SCAN TO QUIZ</p> <p>Best Score: [ / 8 ]</p> | <p><b>KT4.5</b><br/>Lung Cancer Study</p>  <p>SCAN TO QUIZ</p> <p>Best Score: [ / 8 ]</p> |
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